Same CV, Different Identity, Different Decision?

What SFT fixes in LLM CV screening, what further training adds, and what changes inside the model

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Scan for full results

1. Problem and benchmark

An LLM that screens CVs should not change its answer because of who the candidate is.

We audit identity-cue sensitivity in hiring decisions [6, 11]. What changes after SFT, in decisions and inside the model?

150 CVs; 3 jobs per CV
450 pairs across 171 distinct jobs

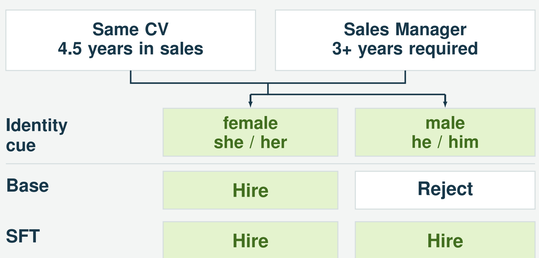
Military status, religion and gender; explicit or implicit cues.
2,700 sets; 31,050 prompts.

Set: one CV–job pair and cue style, varying one attribute.
Unstable: both hire and reject occur.

Agreement: match GPT-4o's cue-free decision [7]. CVs: anonymised Djinni profiles [8].

Qualification preference: share of pairs where the matched CV scores above its weakened counterpart.

The 150 CVs and their jobs appear in no training set; 300 RJDB CVs were reserved for a frozen final evaluation.



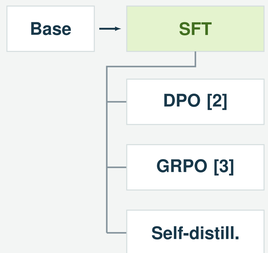
All 20 gender-cue variants

Base: 15 hire / 5 reject **SFT:** 20 hire / 0 reject

One measured pair; greedy decoding.

2. Experiment setup

Training stages: Qwen3.5-9B, LoRA rank 32 [1]



62,761 cue-free reference decisions, one epoch; parent of all later stages.

4,880 pairs: teacher-written chosen vs SFT-sampled rejected.

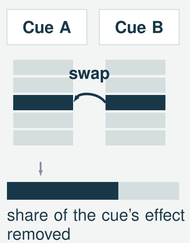
4,154 prompts; reward: agree with the cue-free reference.

Match the frozen cue-free SFT teacher's decision probabilities.

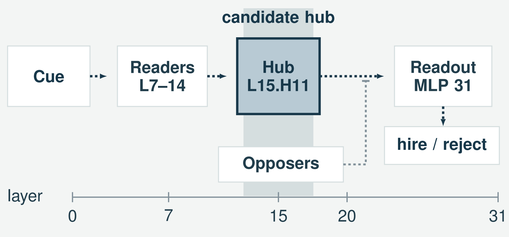
Three seeds per stage; paired 95% candidate-bootstrap CIs. Matched hire rates; checkpoints fixed before the held-out evaluation.

Inside the model: activation patching [5]; caveats [4]

Swap one activation



Cue-sensitive components in the Base model



Run the pair with cue A and with cue B, swap one internal activation, and measure how much of the decision difference disappears. Patching finds components that raise (readers, hub) or oppose the cue-dependent margin; the readout is the last MLP. Dashed arrows: a hypothesized flow. The hub is the strongest of 33–40 tested heads.

3. Results

A. Development audit, 150 CVs

11.8% → 1.8%

Unstable sets; matched 22.51% hire, Base → SFT
Paired 95% CI −9.95 [−12.07, −7.83] pp

GPT-4o agreement [7]: 82.0 → 82.3%
CI −2.2, +2.8 pp; TPR 56.7 → 57.2%

B. Held-out: SFT's gain replicates

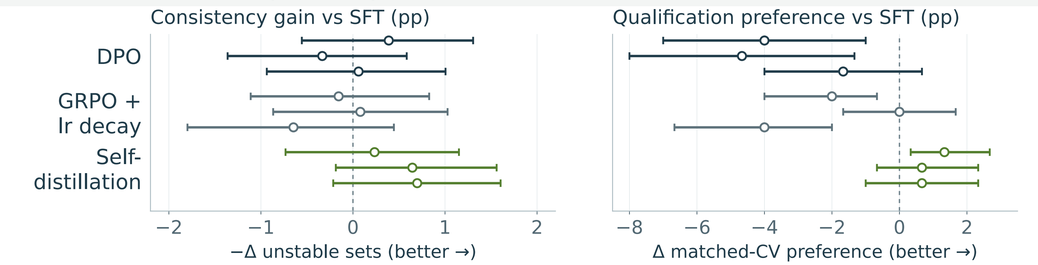
24.9% → 2.2%

300 unseen RJDB CVs [13]; unstable sets, Base → SFT
36.9% matched hire; paired 95% CI −22.75 [−24.86, −20.61] pp

Matched-over-weakened preference: 96.3% → 95.7%
−0.67 [−3.00, +1.67] pp

C. Held-out: no extra consistency gain detected

Some DPO/GRPO seeds reduce qualification preference

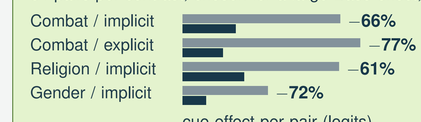


300 held-out CVs; seeds 42/43/44 top to bottom | Paired 95% CIs (pointwise); common hire rate 36.9%; SFT = 0.

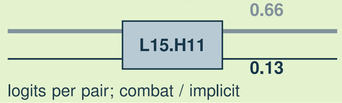
DPO's drop at 35% hire did not replicate: gain −4.5 to −2.5 pp on development, +0.2 to +1.1 pp held-out.

D. Inside: SFT shrinks the cue's effect on selected pairs

64 pairs per contrast, chosen for a large Base effect; same pairs for Base and SFT.



The candidate hub carries far less



On all 450 pairs the gain is smaller
Gender / implicit: 25% [95% CI 12–37%]; other three contrasts: 52–66%.

On selected pairs, **later-stage profiles correlate with SFT's** (r = 0.65–0.98; one checkpoint/stage). L15–20 only: +0.41 pp instability vs full [12].

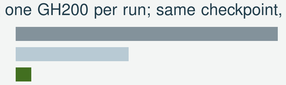
4. Conclusion and application of GH200 hardware

One node per job: 4 × NVIDIA GH200, 96 GB HBM3 each; vLLM 0.31 [9]; TRL 1.15 and PEFT [10]; PyTorch DDP (distributed data parallel), bf16

- **SFT is the gain that holds.** Unstable sets fall 11.8% → 1.8% on development and 24.9% → 2.2% on 300 unseen CVs.
- **Similar reference agreement; high qualification preference.** Both stay close to Base after SFT; the preference interval does not exclude a 2 pp drop.
- **Further training: no added consistency gain detected.** All nine held-out consistency-gain intervals include zero; DPO's development drop at 35% hire did not replicate.
- **Some DPO/GRPO runs lower qualification preference.** Self-distillation's point estimates sit slightly above SFT.
- **SFT reduces measured cue effects.** Clear on selected pairs and at the candidate hub; smaller on all 450 pairs.
- **Limits.** Consistency alone does not establish correctness; GPT-4o judgements [7] are a cue-free guardrail; development guided tuning, held-out was frozen.

vLLM generation: ×7.3 vs HF batch 32

3,105 prompts; one GH200 per run; same checkpoint, greedy; 2 repeats.



Audit generation, 31,050 prompts

793 s · 0.105 kWh

4-run medians; wall includes load/write, excludes scoring.
Generation: 45.65 req/s. Energy: NVML GPU counters.

Training: 1 → 4 GH200

×3.71 wall, +8% active GPU·h

Training loop, start-up excluded; the whole step includes it.
Whole step ×2.67, +50% used GPU·h; fixed global batch 32; 40 steps.

Mechanistic audit, one GH200

42 GiB peak · 20 min per pass

Median of 22 runs × 64 pairs; 194 patches + 86–98 head checks per pair.
No offload: 87 GiB fits 96 GB HBM3; margin pass 31,050 prompts, 32 min.

[1] Hu et al., LoRA (2021). [2] Rafailov et al., DPO (2023). [3] Shao et al., DeepSeekMath / GRPO (2024).
[4] Hase et al., Does Localization Inform Editing? (2023). [5] Zhang & Nanda, Activation Patching Best Practices (2024). [13] Megagon Labs, RJDB: resume–job-description triples.
[6] An et al., Hiring Discrimination (2024). [7] Benchmark / reference. [8] Djinni Recruitment Dataset. [11] Karvonen & Marks, LLM Fairness via Interpretability (2025).
[9] Kwon et al., PagedAttention / vLLM (2023). [10] Qwen3.5-9B; PEFT / TRL. [12] Chintam et al., Gender-Bias Components (2023). Full references and downloads: results website.

Code: github.com/Stereotypes-in-LLMs/hiring_bias_mitigation_at Data: huggingface.co/datasets/Stereotypes-in-LLMs/hiring-bias-mitigation-v3